EXAMINATION NUMBER: _____

**ZOMBA CATHOLIC SECODARY SCHOOL**

**2014-2015 MSCE MOCK EXAMINATIONS**

# MATHEMATICS PAPER I

**(100 marks)**

***Date: Monday, March 16*** ***Time Allowed: 2hours***

## *Instructions*

- ✓ ***Check that your question paper has 9 pages***
- ✓ ***Answer all questions in the spaces provided.***
- ✓ ***All the working must be clearly shown.***
- ✓ ***Electronic calculators may be used.***

1. Simplify $2n\sqrt{n} - n\sqrt{16n} + 5\sqrt{n^3}$ (4 marks)

2. The $8^{th}$ term of an arithmetic progression (AP) is three times as much as the $2^{nd}$ term. If the sum of the first 10 terms of the AP is 325, calculate the first term of the AP. (5 marks)

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3. (a) Make b the subject of the formula bt = $\frac{4c-a}{ab^2}$ + pb (4 marks)

(b) The distance between the points P (-2, 2d ) and Q (d, -14) is 25. Find the possible values of d. (5 marks)

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4. ***Figure 1*** shows a circle ABT in which DTC is a tangent to the circle at T. AB = BT.

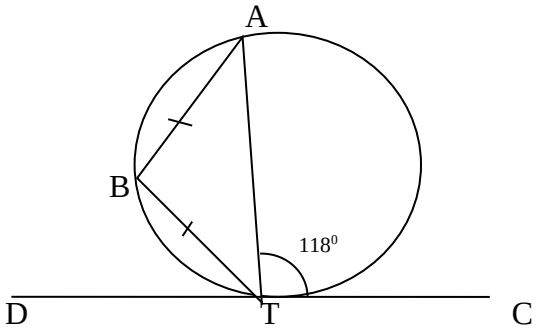


***Figure 1***

If angle ATC = $118^0$, calculate the value of angle BTD. (4 marks)

5. The expression $x(Ax - 3)(Bx + 5)$ is identical to $8x^3 + 14x^2 + Cx$ for all values of x. Find the values of A, B and C given that they are all whole numbers. (4 marks)

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6. ***Figure 2*** is a speed time graph of an object

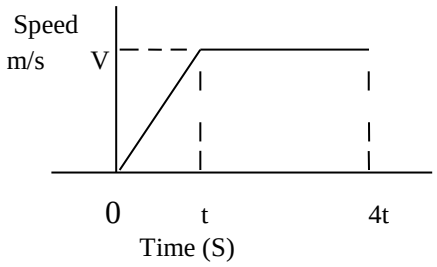


***Figure 2***

Given that the total distance travelled is 1400 m, and the acceleration is $4mls^2$ for the first t seconds, find:

(a) the value of v

(b) the value of t (5 marks)

7. In a group of 50 form 4 learners, (x +1) students take Maths, Biology and Chemistry. 9 take Biology and Chemistry only, 8 take Maths and Biology only and 5 take Maths and Chemistry only. If x students take Maths only, x + 3 students take Chemistry only and x students take Biology only; by using a venn diagram, calculate the number of students who take Maths. (5 marks)

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8. (a) A chord is 12cm from the centre of a circle of radius 13cm. Calculate the length of the chord. (4 marks)

(b) Without using a calculator, simplify $\frac{log\ 81 - log\ 9}{\log 81 - log\frac{1}{9}}$ (4 marks)

9. ***Figure 3*** shows a cup shaped parabola f(x).

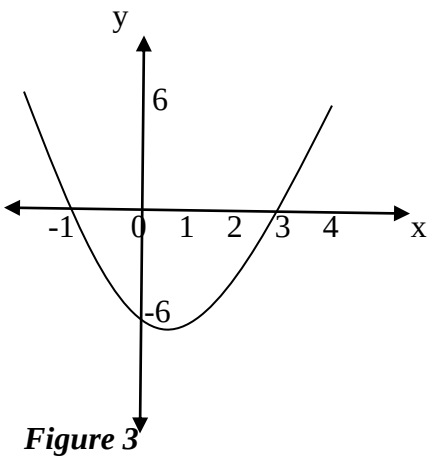


***Figure 3***

Write down the equation for the (i) function f(x) (3 marks)

(ii) line of symmetry (2 marks)

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10. (a) Express $\frac{2}{m+1}$ - $\frac{m+3}{2+3m+m^2}$ as a single fraction in its simplest form. (5 marks)

(b) Given a function $h(x) = w^2x + 30$ if $h(-3) = 3$, find the value of w. (4 marks)

11. Given that r varies partly as d and partly inversely as d. The value of r = 7 when d = 2 and r = 5 when d = 1. When r = 12.5, find the possible values of d. (5 marks)

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12. A fair coin is tossed and a fair die is rolled respectively, find the probability of obtaining a tail and a five. (3 marks)

13. Calculate the standard deviation of 4, 8 and 3. (5 marks)

14. In a quadrilateral ABCD, $\overrightarrow{BA} = 2\underline{f}$, $\overrightarrow{CB} = \underline{e}$ and $\overrightarrow{CD} = 3\underline{f}$. Find $\overrightarrow{AD}$ in terms of $\underline{e}$ and $\underline{f}$. (5 marks)

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15. ***Figure 4*** shows a triangle PQR such that angle PQR = $60^0$, angle QRP = $45^0$ and PQ = 14cm.

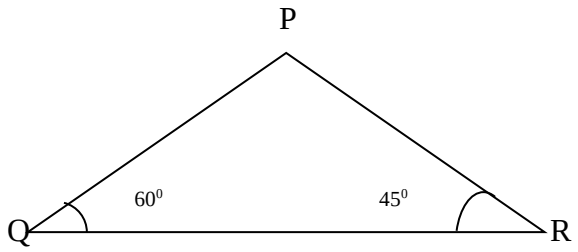


***Figure 4***

Calculate the length of PR, leaving your answer in simplified surd form. (4 marks)

16. The corresponding bases of two similar triangles are in the ratio 4**:**7. If the area of a smaller triangle is $96m^2$, calculate the area of the bigger triangle. (4 marks)

17. Given that matrix T = $\begin{pmatrix} 3 & -5 \\ 1 & 2 \end{pmatrix}$; if $T^2 + K = 3T$, find matrix K. (5 marks)

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18. ***Figure 5*** shows a region C that is bounded by three inequalities.

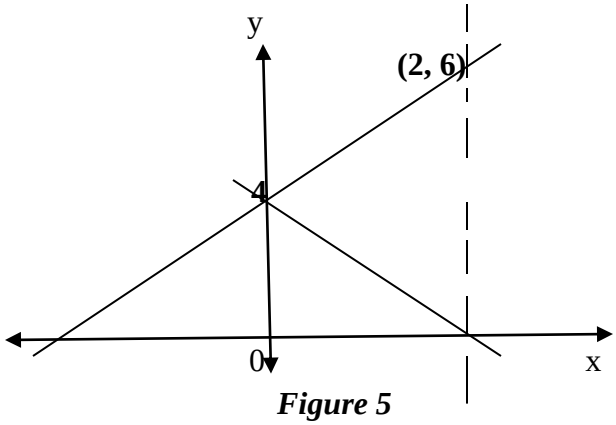


***Figure 5***

Write down three inequalities that describe the region C. (3 marks)

19. Solve the simultaneous equations (5 marks)

$p^2 - pr - 3 = 0$

$2r = p + 1$

20. A point T (-5, 8) is translated 2 units to the left and 6 units up to the point T ′. Calculate the coordinates of T '. (3 marks)